Yash Jaiswal

+91-6261306890 | yash.r.jaiswal2000@gmail.com | LinkedIn | GitHub | Coding Profiles

Profile Summary

Software engineer skilled in **backend development, distributed systems, system design, and AI-driven applications**, with experience building **scalable, reliable software** using modern technologies.

Education

National Institute of Technology Jamshedpur — Jamshedpur, Jharkhand
Master of Technology, Data Science and Engineering | CGPA:8.9 — *Aug. 2024 – May 2026*
Coursework: Machine Learning, Deep Learning, Big Data Analytics, Natural Language Processing, etc.

Work Experience

SDE Intern — August 2025 – May 2026
Sanas.AI — *Bangalore, Karnataka*

- **Tech Stack: Go, NestJS, .NET Core, React, ClickHouse, AWS Aurora PostgreSQL**
- Contributed to a centralized **Authentication Service** in Go integrating **Ory Kratos, Hydra, and Keto**, working on selected authentication and authorization flows involving **OAuth 2.0/OIDC**, OTP-based authentication, JWTs, and **ReBAC**.
- Proposed and architected a real-time **license and usage tracking system** across a 3-tier SaaS account hierarchy (Account → Groups → Subgroups), enabling customers to monitor live seat consumption and entitlements at any level of the org tree — directly tied to billing accuracy and revenue protection.
- Designed the **backend query engine** powering the live usage dashboard, handling recursive aggregation across arbitrary subgroup depths with pruning on locked subtrees; the same engine backs planned daily usage graphs with full timezone-aware timestamps.
- Developed full-stack expertise across **two generations of the portal**, resolving **100+ bugs** with end-to-end ownership spanning React frontend, NestJS handlers, service layers, DTOs, and Aurora PostgreSQL — including ongoing critical path work and security hardening (SQL injection prevention, XSS sanitization via systematic input validation).

Technical Skills

- **Languages**: Go, Java, Python, C/C++, JavaScript, TypeScript, SQL
- **Backend**: Node.js, Express.js, NestJS, FastAPI, RESTful APIs, Kafka, Gin, SpringBoot
- **Databases**: ClickHouse, PostgreSQL (Aurora), MySQL, MongoDB, Redis, DynamoDB
- **Cloud & DevOps**: AWS EC2, Docker, Kubernetes, Nginx, CI/CD, GitHub Actions
- **Tools & Data Science**: Git, Linux, Postman, New Relic, Pandas, NumPy, Scikit-learn, Matplotlib
- **Core Competencies**: Data Structures & Algorithms, System Design, Computer Networks, Operating Systems

Projects

Samvada AI- Real-Time Voice AI Platform & SaaS Portal — Sept 2026

- Tech Stack: **Go, Gin, React, TypeScript, PostgreSQL, pgvector, ClickHouse, Redis, Apache Kafka, Docker, WebSockets, gRPC, REST APIs, Ory (Kratos, Hydra, Keto), ReBAC, Twilio, Plivo, Deepgram, FishAudio**.
- Architected a full-duplex, streaming voice pipeline in **Go** connecting PSTN networks (**Plivo, Twilio**) with **Deepgram** (STT), **LLM token streaming**, and **FishAudio** (TTS), achieving **1.19s end-to-end response latency**.
- Engineered human-like conversational realism using instant fillers (**FillerService**), active listening cues (**BackchannelService**), background room noise (**AmbienceService**), < 10 **ms Barge-In** interruption, and a **pgvector-powered Hybrid RAG engine** for low-latency knowledge retrieval.
- Developed a full-stack SaaS portal using **React** and **Go (Gin)** to configure AI voice agents, monitor live calls through real-time audio visualizations, and manage SIP phone number provisioning.
- Developed backend microservices in **Go (Gin)**, integrating open-source **Ory** services for JWT/OTP-based authentication and fine-grained **ReBAC** authorization, alongside an **Application & Billing Service (ABS)** for multi-tenant workspace management and wallet deductions using **PostgreSQL**, supported by a dedicated background worker.
- Implemented an event-driven **OLAP Reporting Service** with a dedicated background worker to process call telemetry into **ClickHouse**, enabling low-latency analytical queries and usage dashboards; containerized microservices, workers, and databases using **Docker**.

Achievement

- Solved **2500+ DSA Problems** on different platforms like Leetcode, CodeForces, GFG, Coding Ninjas, etc. Profiles
- **Max Rating: 1514** in contest rating of **Codeforces (Specialist)**. Link
- **Top 4%** in contest rating of **Leetcode (Knight)**. Link
- **Top 1%** in contest of **GeeksForGeeks (4 Stars)**. Link
- **Top 3%** in contests of **Coding Ninjas (Master)**. Link.